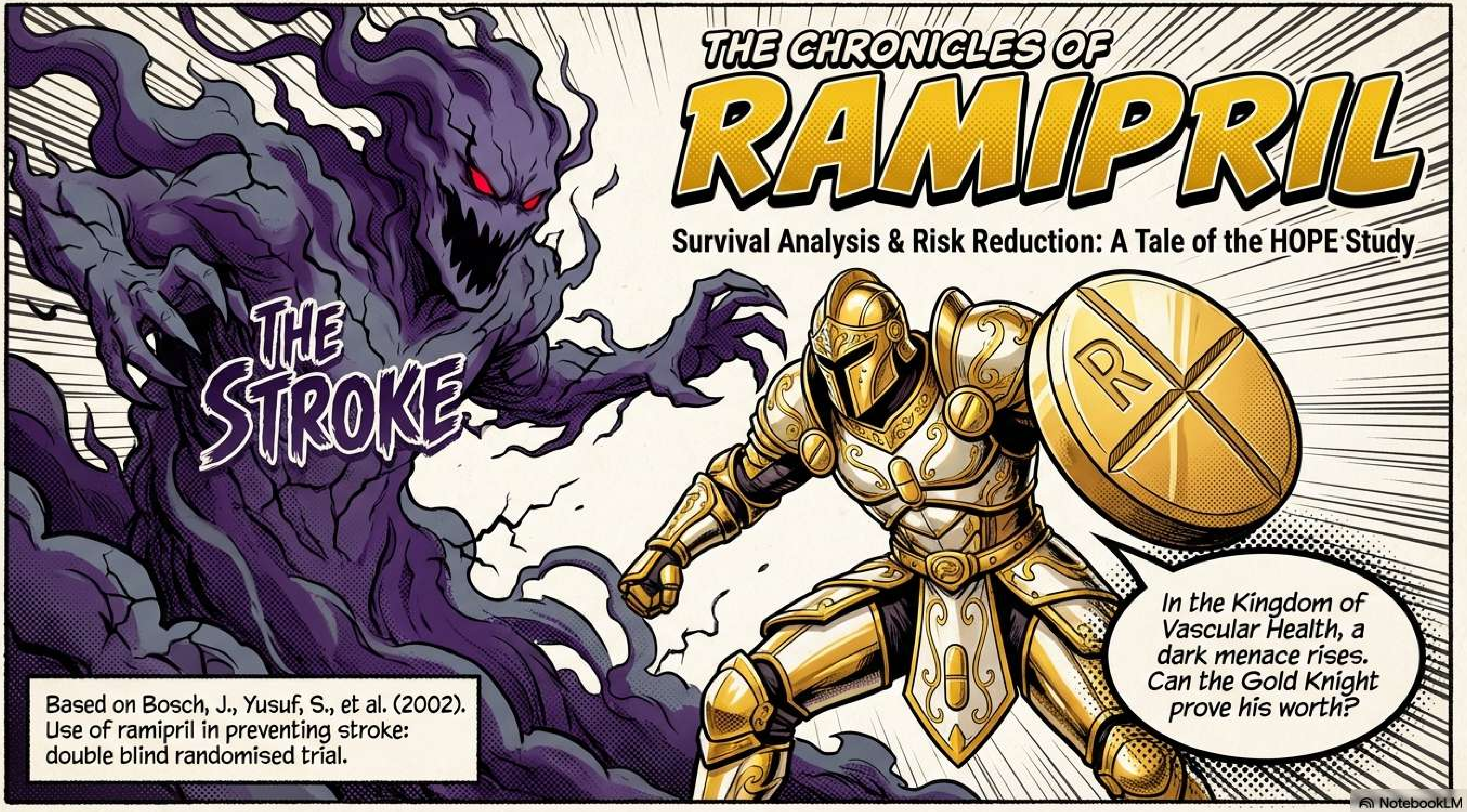
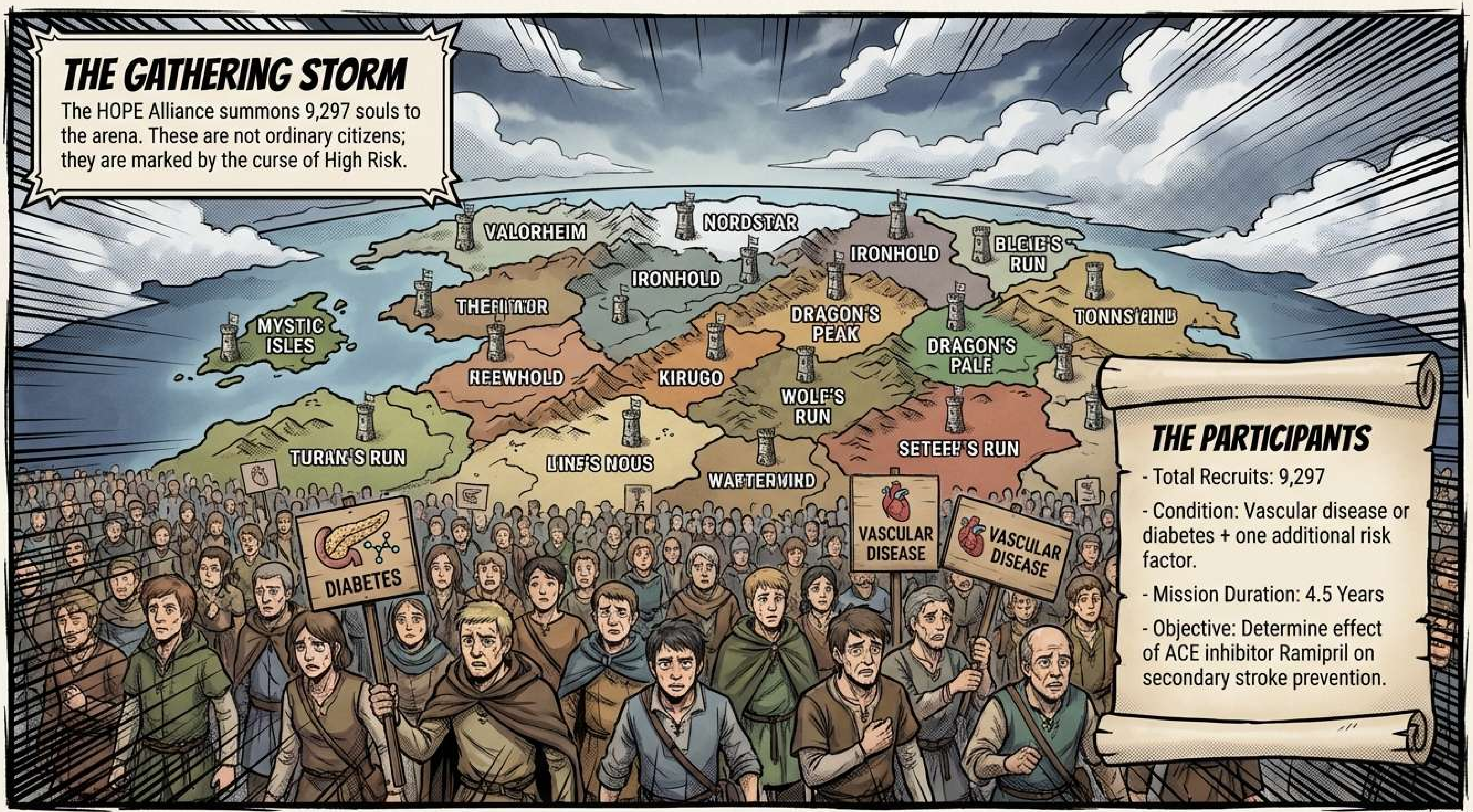
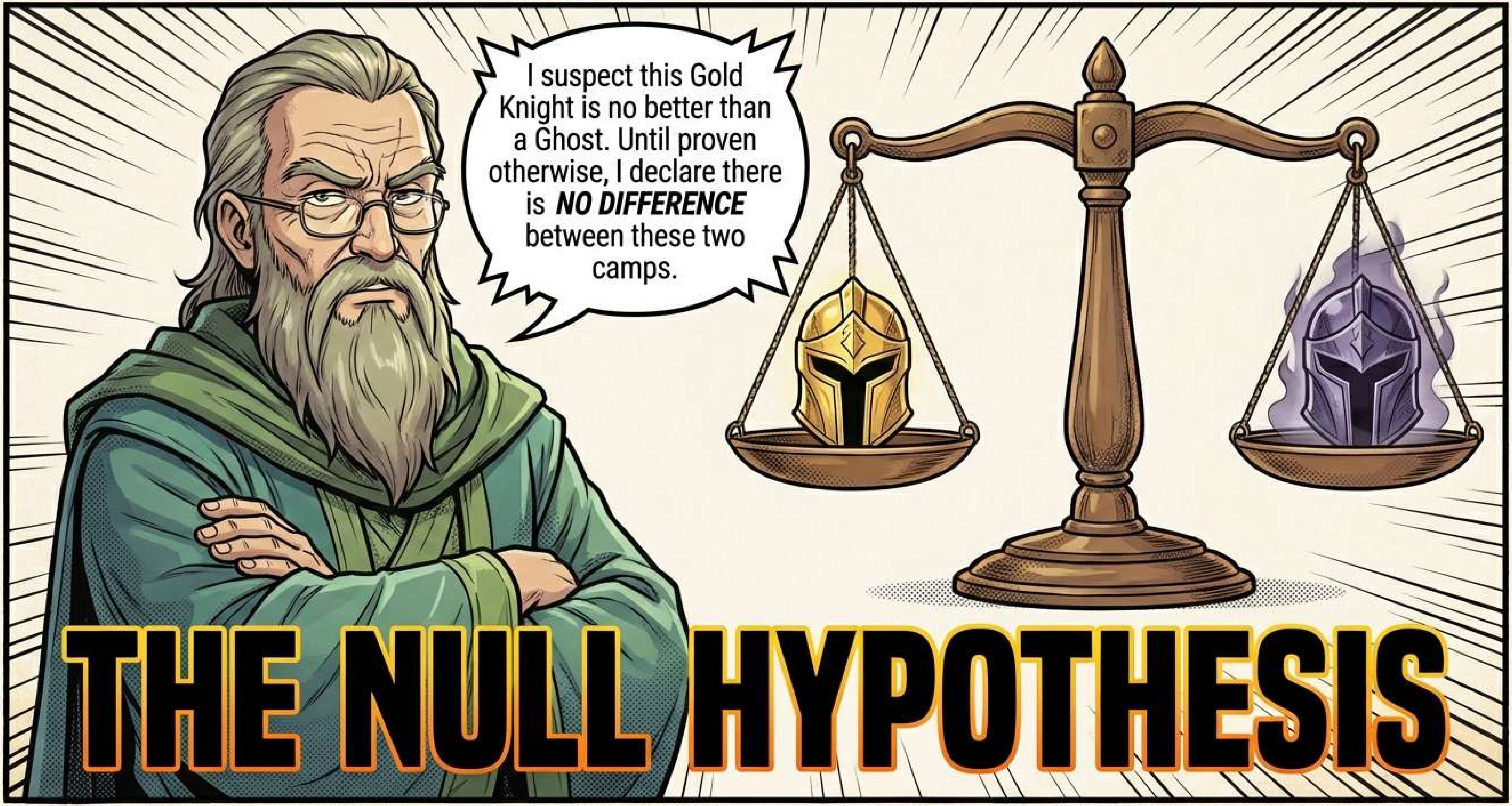

THE CHRONICLES OF
RAMIPRIL
Survival Analysis & Risk Reduction: A Tale of the HOPE Study
THE STROKE
R
In the Kingdom of Vascular Health, a dark menace rises. Can the Gold Knight prove his worth?
Based on Bosch, J., Yusuf, S., et al. (2002). Use of ramipril in preventing stroke: double blind randomised trial.

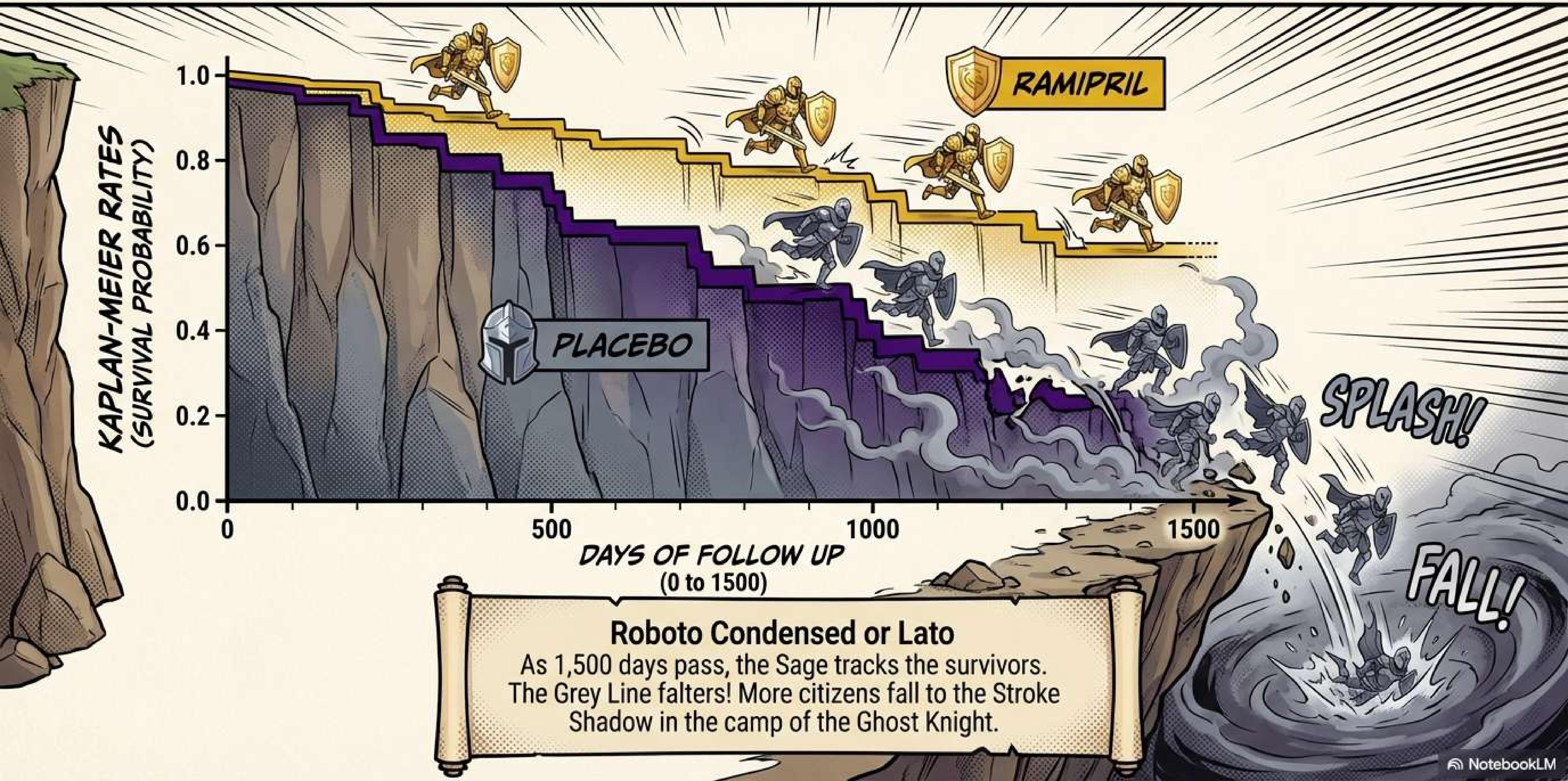

THE GATHERING STORM
The HOPE Alliance summons 9,297 souls to the arena. These are not ordinary citizens; they are marked by the curse of High Risk.
VALORHEIM
NORDSTAR
IRONHOLD
IRONHOLD
MYSTIC ISLES
DRAGON'S PEAK
KIRUGO
WOLF'S RUN
TURAN'S RUN
DIABETES
VASCULAR DISEASE
VASCULAR DISEASE
THE PARTICIPANTS
- Total Recruits: 9,297
- Condition: Vascular disease or diabetes + one additional risk factor.
- Mission Duration: 4.5 Years
- Objective: Determine effect of ACE inhibitor Ramipril on secondary stroke prevention.

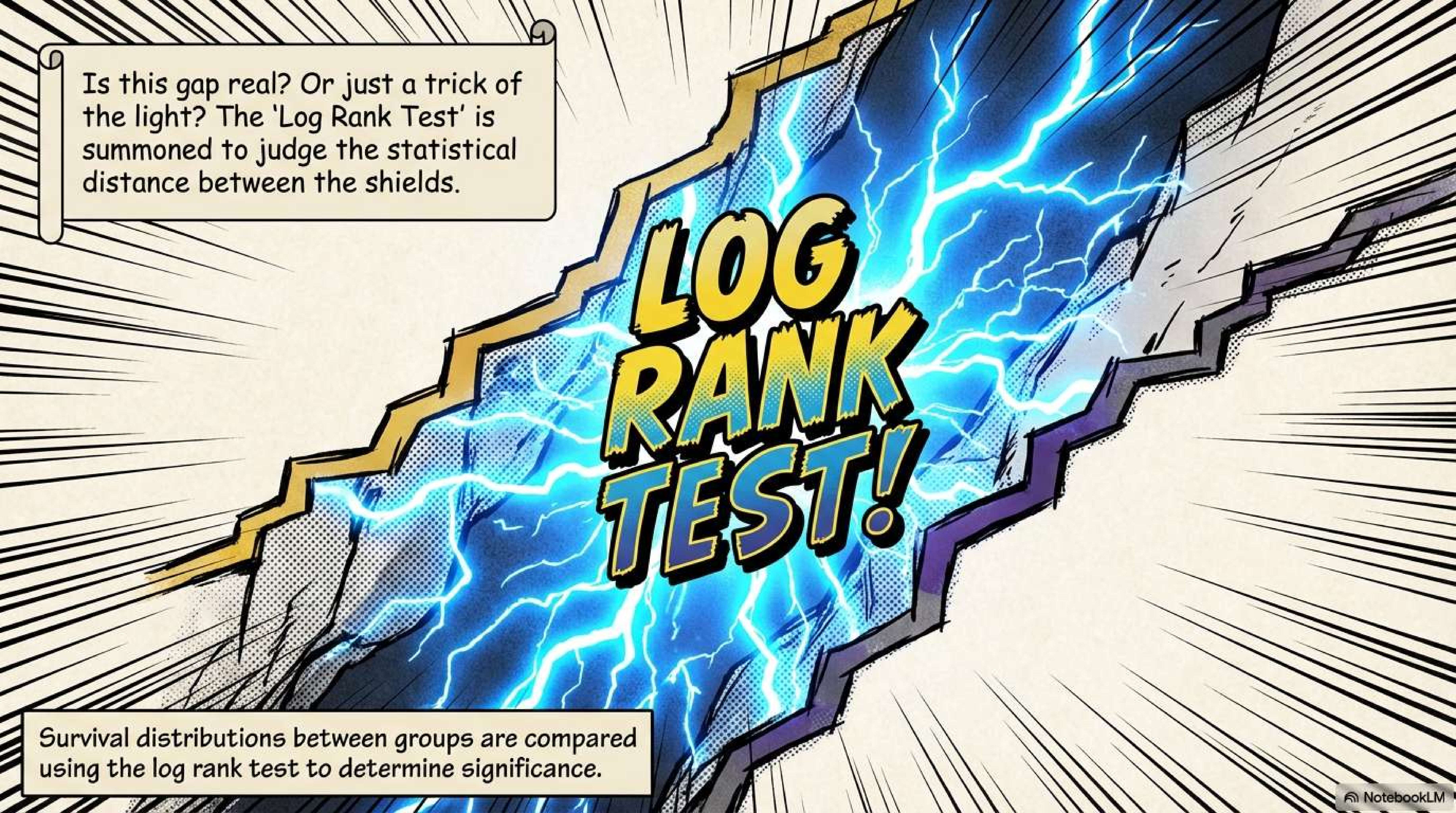

RANDOMIZATION
RAMIPRIL GROUP
(n=4645)
PLACEBO GROUP
(n=4652)
THE DOUBLE-BLIND SPELL.
To test the true power of the shield, fate divides the people.
A fog descends—neither the villagers nor the healers
know who holds the true medicine.

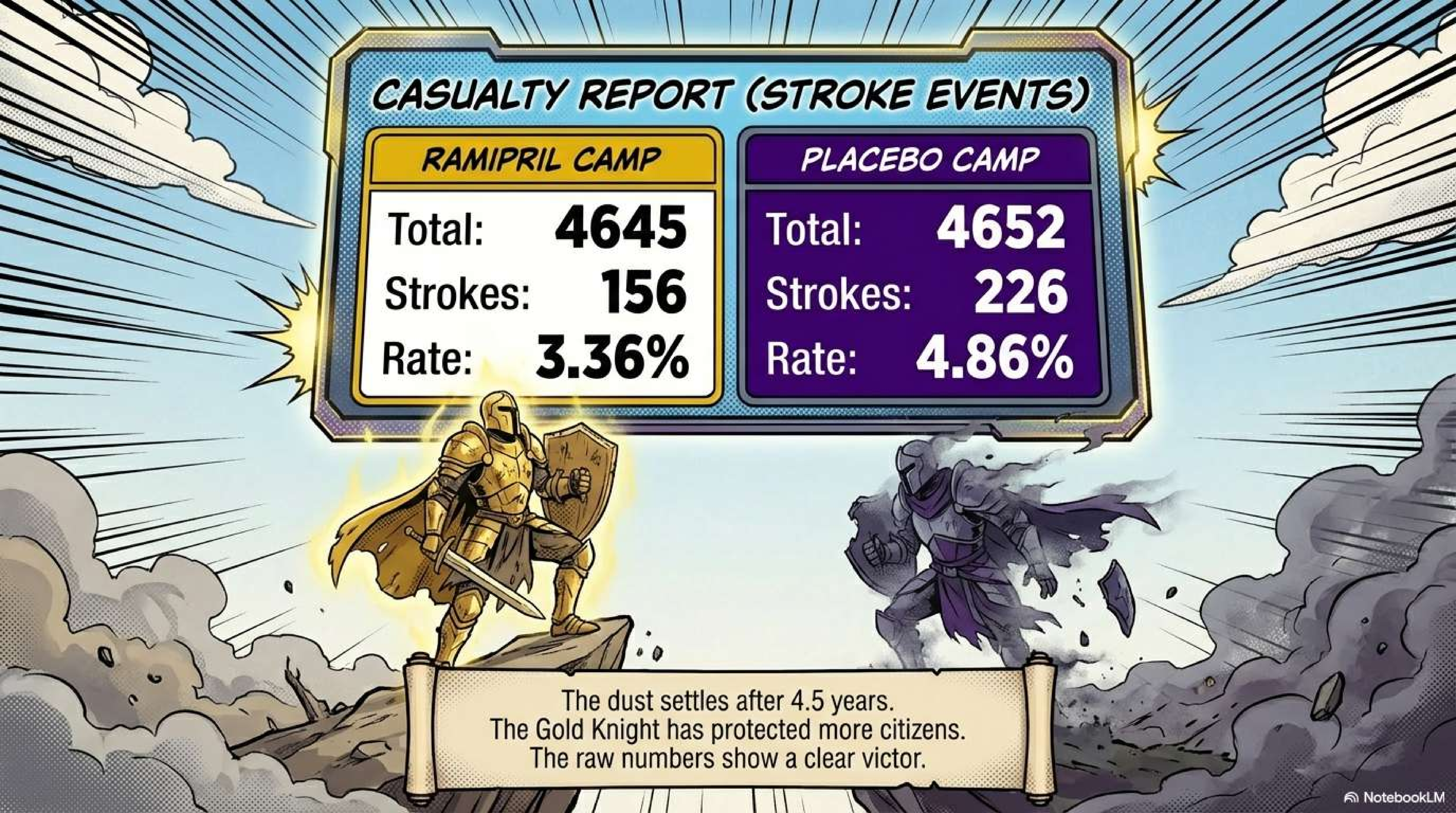


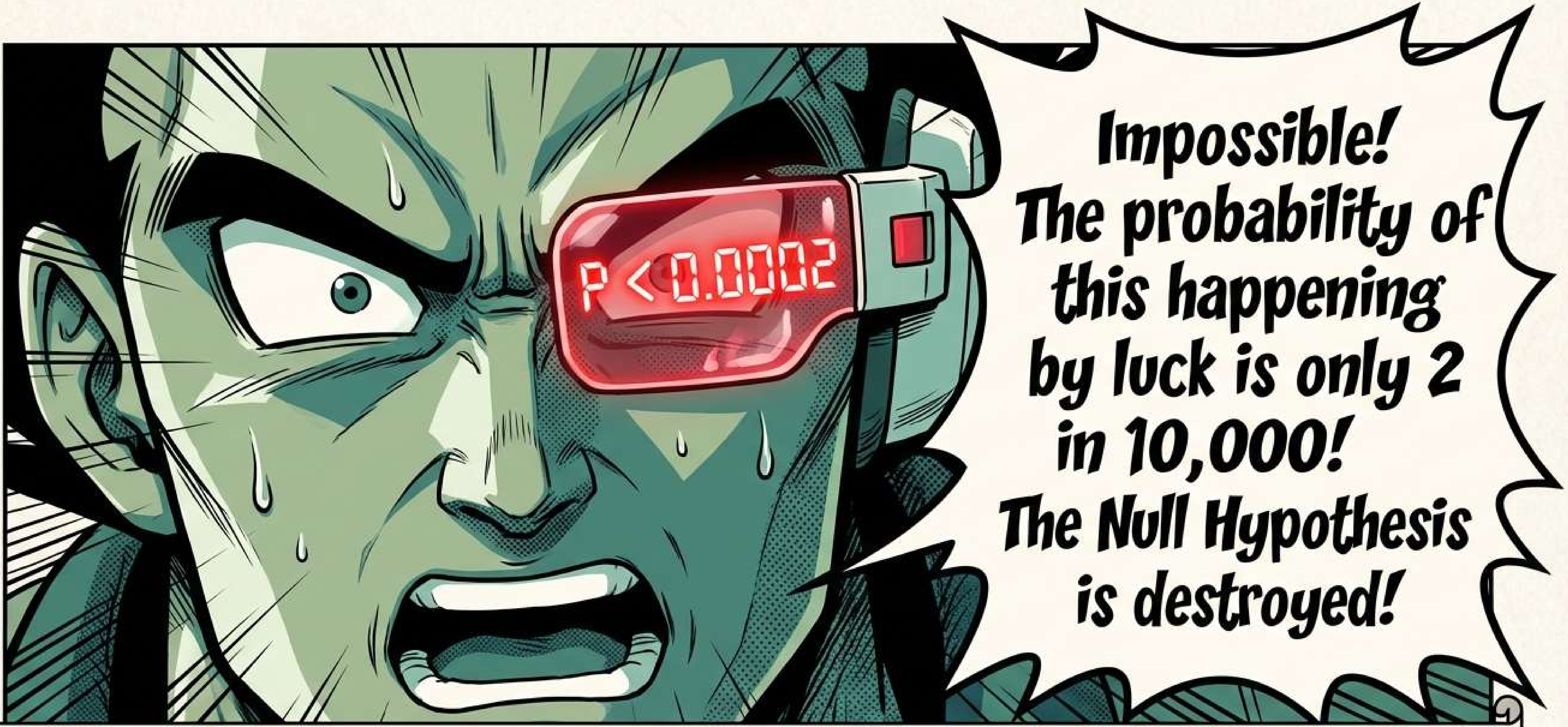
The burden of proof lies with the hero. If the difference in survival is small, the Null Hypothesis stands undefeated.

THE BATTLE OF THE CLIFF (KAPLAN-MEIER PLOT)



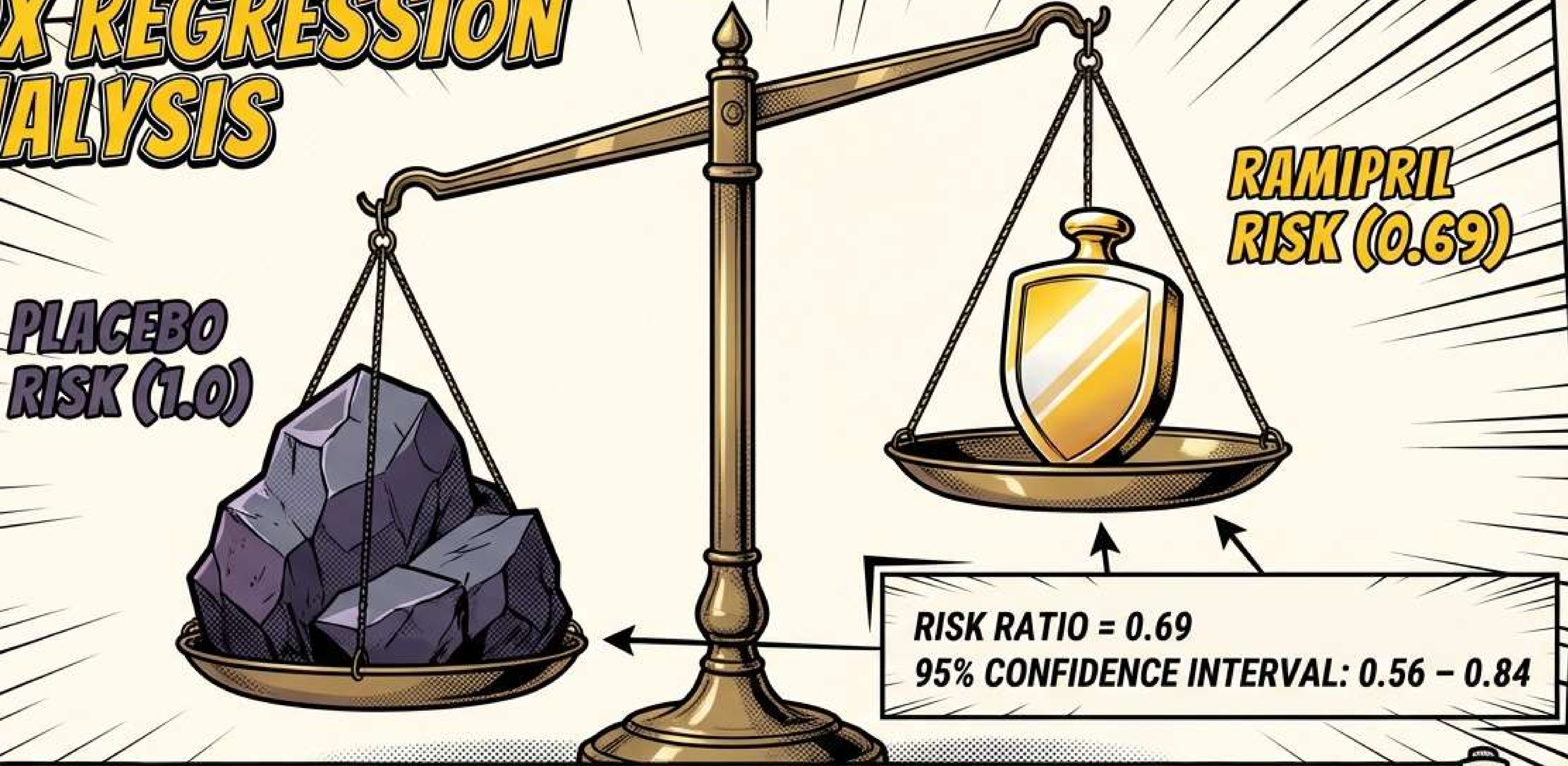

Is this gap real? Or just a trick of the light? The 'Log Rank Test' is summoned to judge the statistical distance between the shields.
LOG RANK TEST!
Survival distributions between groups are compared using the log rank test to determine significance.
NotebookLM


CASUALTY REPORT (STROKE EVENTS)
RAMIPRIL CAMP
Total: 4645
Strokes: 156
Rate: 3.36%
PLACEBO CAMP
Total: 4652
Strokes: 226
Rate: 4.86%
The dust settles after 4.5 years.
The Gold Knight has protected more citizens.
The raw numbers show a clear victor.
NotebookLM



Significance Level: A P-value of 0.0002 means the result is highly significant. It is extremely unlikely to be due to chance.

COX REGRESSION ANALYSIS



The Cox Regression technique analyzes the “Hazard.” It reveals that the risk in the Gold Camp is only 0.69 times the risk in the Ghost Camp. Since the Confidence Interval does not cross 1.0, the victory is real.

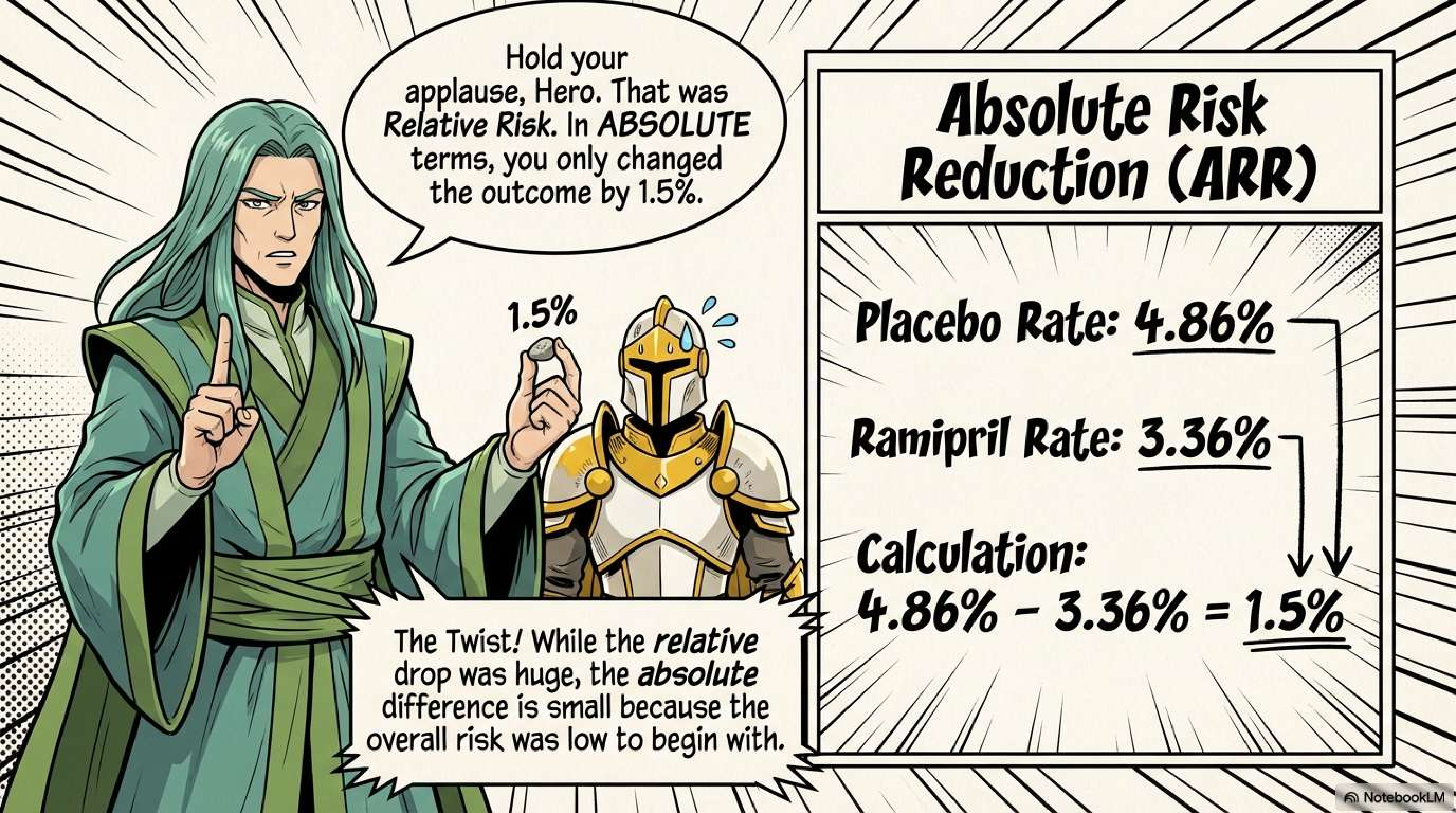
Calculating RRR (Relative Risk Reduction):

Formula: (Placebo Rate – Ramipril Rate) / Placebo Rate

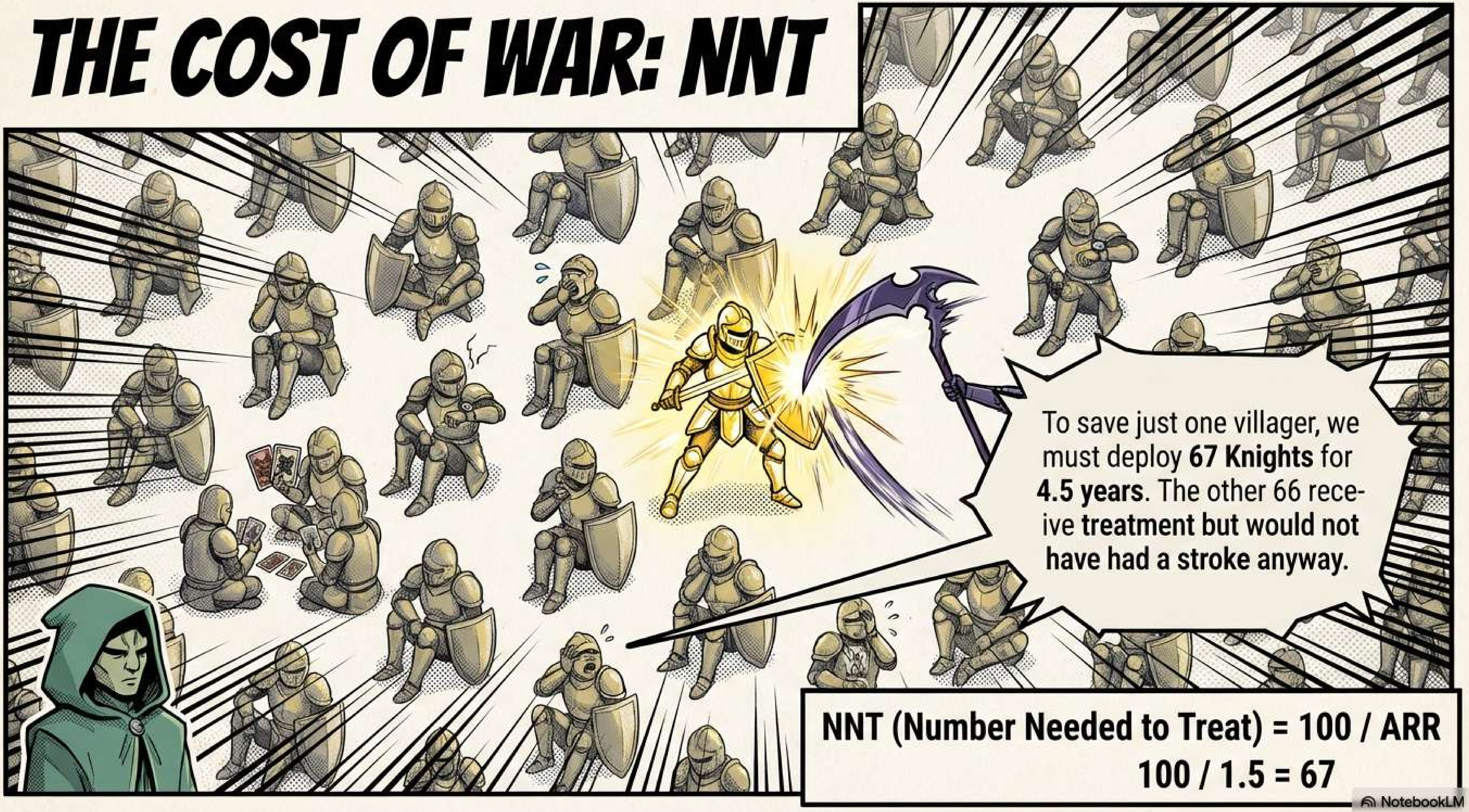
Math: (4.86 - 3.36) / 4.86 = 0.31

Result: 31% RRR

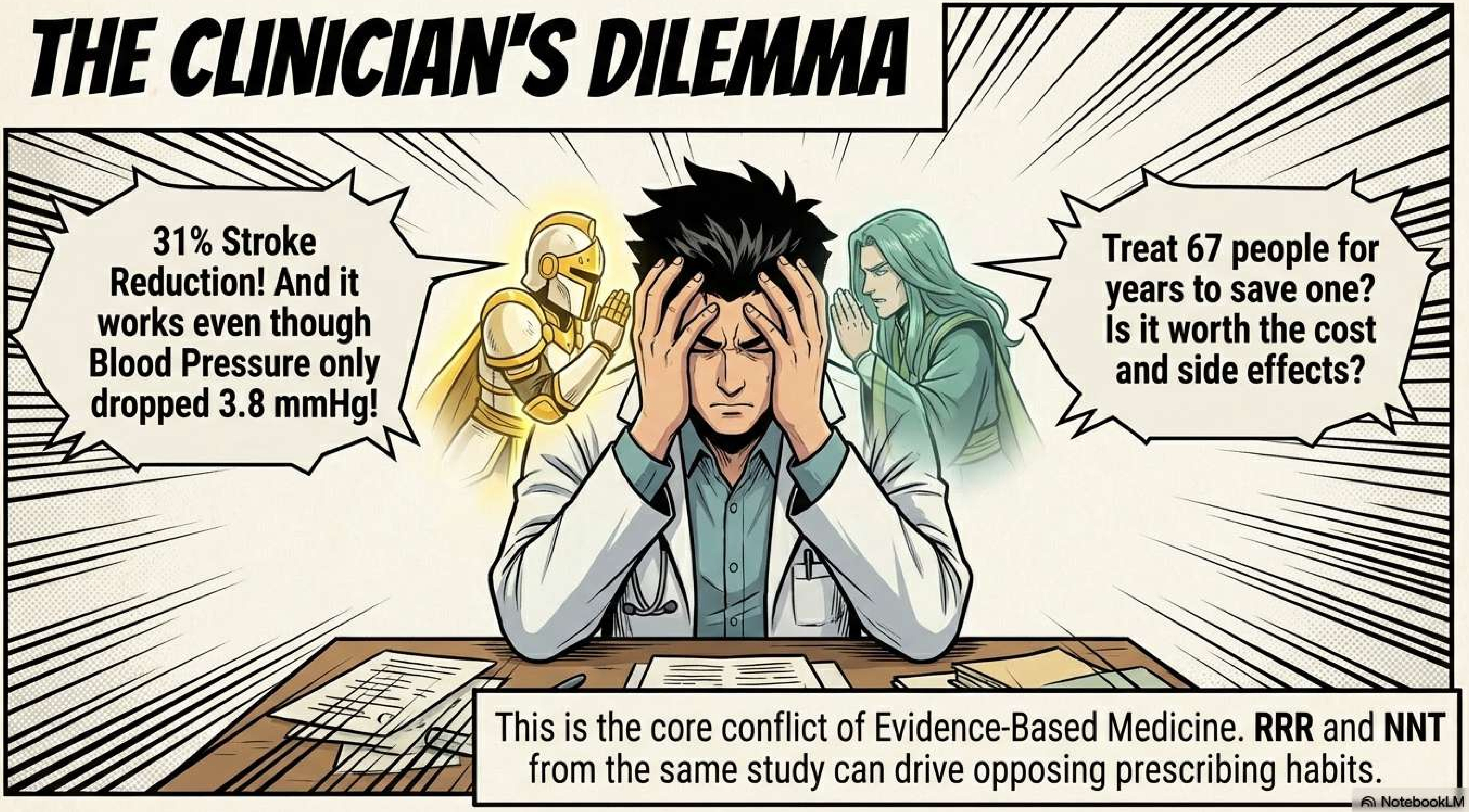

FATAL STROKE STATS
Ramipril Deaths: 17 (0.37%)
Placebo Deaths: 44 (0.95%)
Risk Ratio: 0.39
FATAL STROKE RRR = 61%
Against the deadliest blow, the shield is strongest.
The relative risk of fatal stroke was reduced by 61%.
NotebookLM


Hold your applause, Hero. That was *Relative Risk*. In ABSOLUTE terms, you only changed the outcome by 1.5%.
1.5%
Absolute Risk Reduction (ARR)
Placebo Rate: 4.86%
Ramipril Rate: 3.36%
Calculation:
4.86% - 3.36% = 1.5%
The Twist! While the *relative* drop was huge, the *absolute* difference is small because the overall risk was low to begin with.

THE COST OF WAR: NNT



THE CLINICIAN'S DILEMMA



THE LEGEND RECAP:

1. **Hypothesis:** Ramipril prevents stroke in high-risk patients.
2. **Analysis:** Kaplan-Meier curves & Log Rank Test showed significant survival difference ($P < 0.0002$).
3. **Victory**: Relative Risk Reduction (RRR) = 31%.
4. **Reality**: Absolute Risk Reduction (ARR) = 1.5%.
5. **Cost**: NNT = 67.

Full Citation: Bosch, J., Yusuf, S., Pogue, J., Sleight, P., Lonn, E. et al. (2002). Use of ramipril in preventing stroke: double blind randomised trial. **BMJ 324:** 699–702.